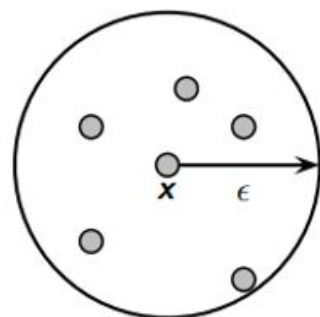
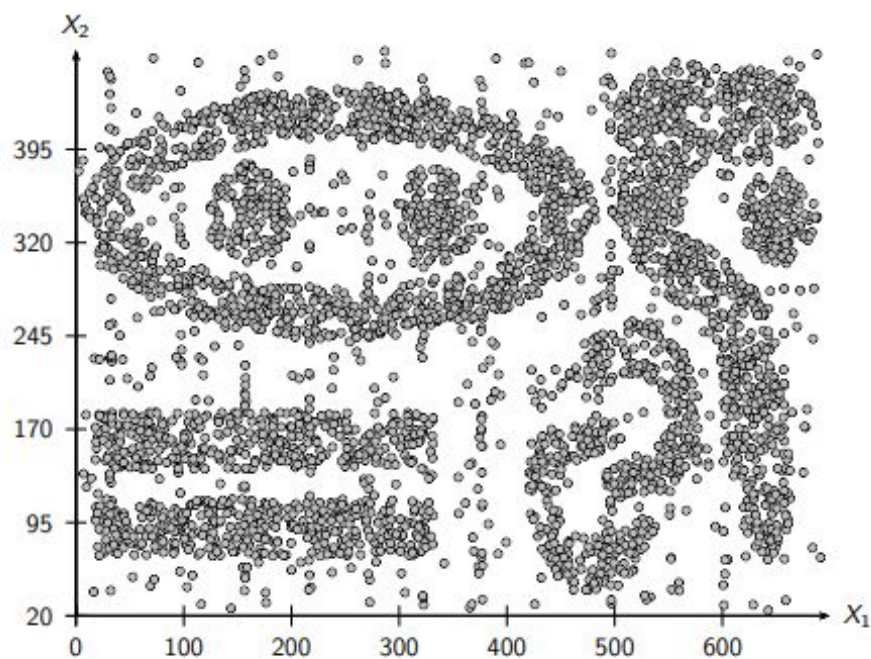


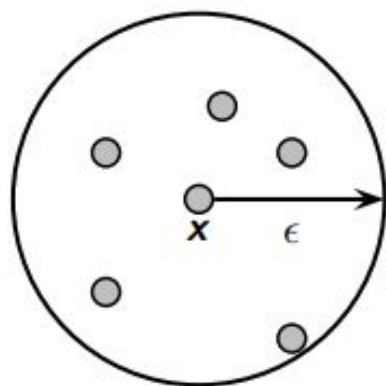
Density- Based Clustering: Questions

- What is the advantage of DBC?
- What is DBC approach?
- Define Core, Border and Noise Points.
- Explain DBSCAN in Brief.
- Write down the DBSCAN algorithm.

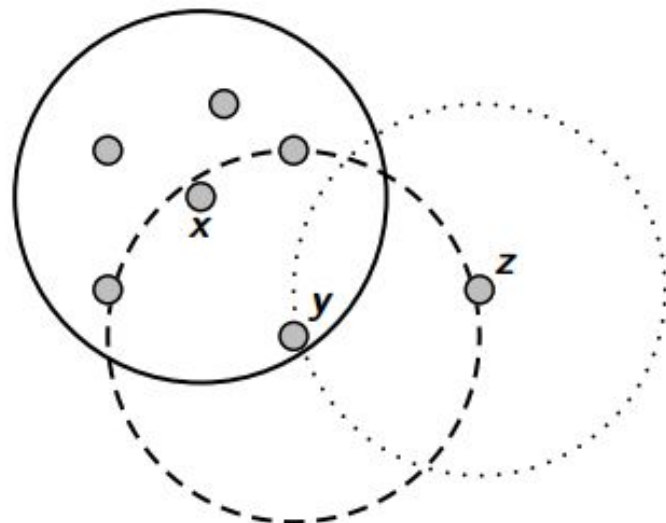
Density-based methods are able to mine nonconvex clusters, where distance-based methods may have difficulty.



(a) Neighborhood of a Point



(a) Neighborhood of a Point



(b) Core, Border, and Noise Points

Define a ball of radius ϵ around a point $\mathbf{x} \in \mathbb{R}^d$, called the ϵ -neighborhood of $\mathbf{x}$:

$$N_{\epsilon}(\mathbf{x}) = B_d(\mathbf{x}, \epsilon) = \{\mathbf{y} \mid \delta(\mathbf{x}, \mathbf{y}) \leq \epsilon\}$$

Here $\delta(\mathbf{x}, \mathbf{y})$ represents the distance between points $\mathbf{x}$ and $\mathbf{y}$. which is usually assumed to be the Euclidean

We say that $\mathbf{x}$ is a *core point* if there are at least *minpts* points in its ϵ -neighborhood, i.e., if $|N_{\epsilon}(\mathbf{x})| \geq \text{minpts}$.

A *border point* does not meet the *minpts* threshold, i.e., $|N_{\epsilon}(\mathbf{x})| < \text{minpts}$, but it belongs to the ϵ -neighborhood of some core point $\mathbf{z}$, that is, $\mathbf{x} \in N_{\epsilon}(\mathbf{z})$.

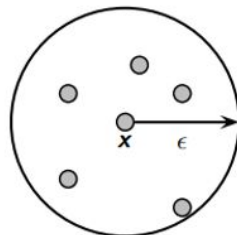
If a point is neither a core nor a border point, then it is called a *noise point* or an outlier.

A point x is *directly density reachable* from another point y if $x \in N_\epsilon(y)$ and y is a core point.

A point x is *density reachable* from y if there exists a chain of points, $x_0, x_1, \dots, x_l$, such that $x = x_0$ and $y = x_l$, and x_i is directly density reachable from x_{i-1} for all $i = 1, \dots, l$. In other words, there is set of core points leading from y to x .

Two points x and y are *density connected* if there exists a core point z , such that both x and y are density reachable from z .

A *density-based cluster* is defined as a maximal set of density connected points.



(a) Neighborhood of a Point

DBSCAN computes the ϵ -neighborhood $N_\epsilon(\mathbf{x}_i)$ for each point $\mathbf{x}_i$ in the dataset $\mathbf{D}$, and checks if it is a core point. It also sets the cluster id $id(\mathbf{x}_i) = \emptyset$ for all points, indicating that they are not assigned to any cluster.

Starting from each unassigned core point, the method recursively finds all its density connected points, which are assigned to the same cluster.

Some border point may be reachable from core points in more than one cluster; they may either be arbitrarily assigned to one of the clusters or to all of them (if overlapping clusters are allowed).

Those points that do not belong to any cluster are treated as outliers or noise.

Each DBSCAN cluster is a maximal connected component over the core point graph.

DBSCAN is sensitive to the choice of ϵ , in particular if clusters have different densities. The overall complexity of DBSCAN is $O(n^2)$.

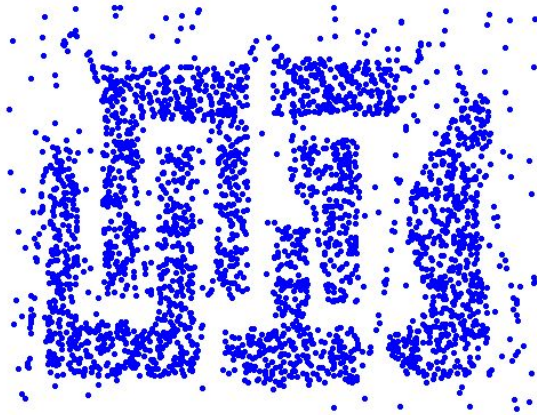
dbscan ($D, \epsilon, minpts$):

```
1  $Core \leftarrow \emptyset$ 
2 foreach  $x_i \in D$  do // Find the core points
3   Compute  $N_\epsilon(x_i)$ 
4    $id(x_i) \leftarrow \emptyset$  // cluster id for  $x_i$ 
5   if  $N_\epsilon(x_i) \geq minpts$  then  $Core \leftarrow Core \cup \{x_i\}$ 
6  $k \leftarrow 0$  // cluster id
7
8 foreach  $x_i \in Core$ , such that  $id(x_i) = \emptyset$  do
9    $k \leftarrow k + 1$ 
10   $id(x_i) \leftarrow k$  // assign  $x_i$  to cluster id  $k$ 
11  DensityConnected ( $x_i, k$ )
12  $\mathcal{C} \leftarrow \{C_i\}_{i=1}^k$ , where  $C_i \leftarrow \{x \in D \mid id(x) = i\}$ 
13  $Noise \leftarrow \{x \in D \mid id(x) = \emptyset\}$ 
14  $Border \leftarrow D \setminus \{Core \cup Noise\}$ 
15 return  $\mathcal{C}, Core, Border, Noise$ 
```

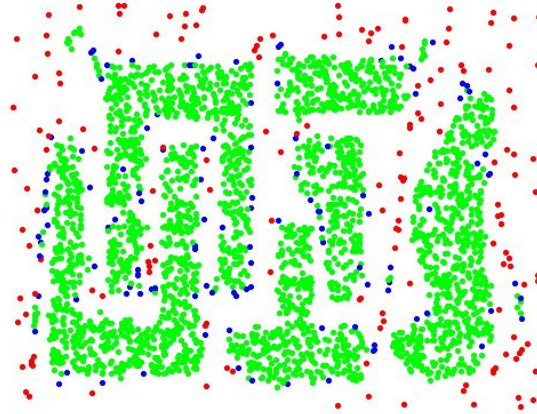
DensityConnected (x, k):

```
16 foreach  $y \in N_\epsilon(x)$  do
17    $id(y) \leftarrow k$  // assign  $y$  to cluster id  $k$ 
18
19   if  $y \in Core$  then DensityConnected ( $y, k$ )
20
```

DBSCAN: Core, Border and Noise Points



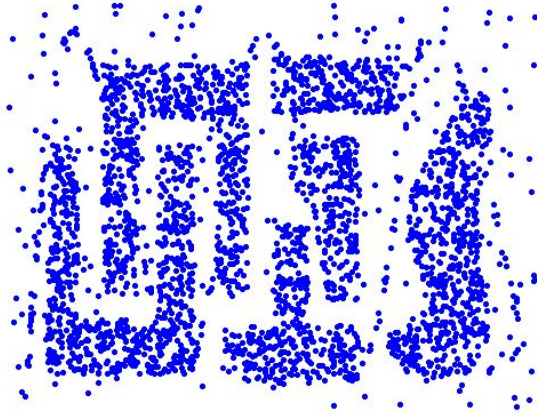
Original Points



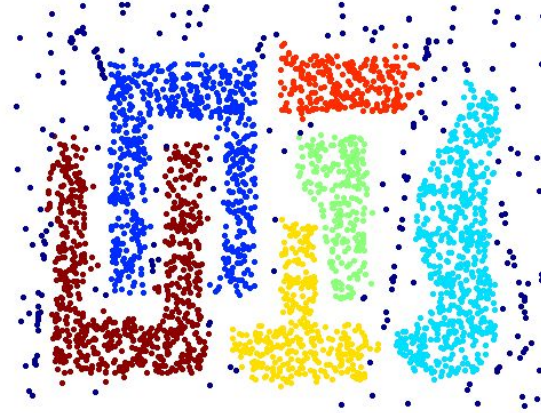
Point types: core,
border and noise

Eps = 10, MinPts = 4

When DBSCAN Works Well



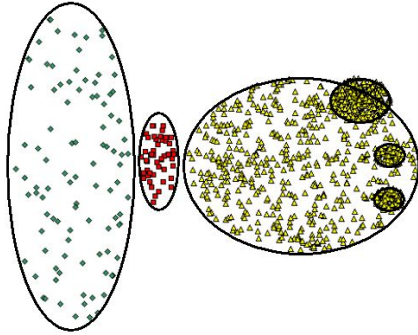
Original Points



Clusters

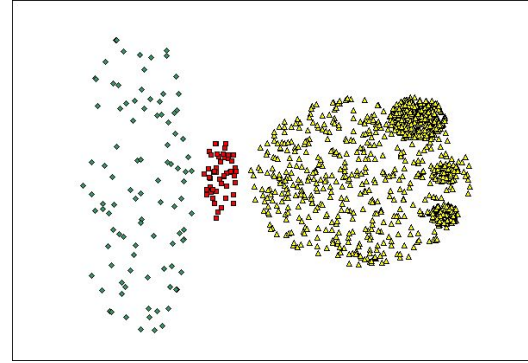
- Resistant to Noise
- Can handle clusters of different shapes and sizes

When DBSCAN Does NOT Work Well

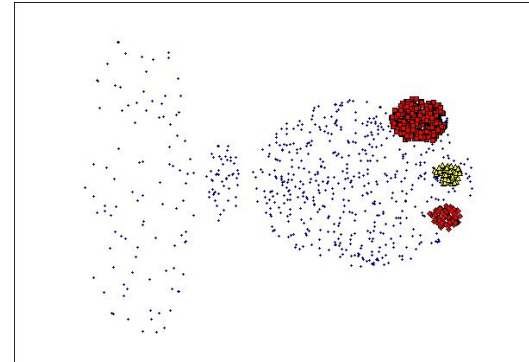


Original Points

- Varying densities
- High-dimensional data



(MinPts=4, Eps=9.75).



(MinPts=4, Eps=9.92)